
Kairos: A Self-Evolving Agent for Formal Verification of Scientific Claims

Anonymous Authors¹

Abstract

We present KAIROS, a self-evolving agent pipeline that formally verifies scientific claims by iterating through refutation, correction, and re-verification. Applied to dopamine-dependent credit assignment in neuroscience, KAIROS formally refutes the two-timescale actor-critic convergence theorem in its commonly-cited form: without a bound on the feature map, the actor iterate diverges under Robbins–Monro step sizes. The corrected statement under summability hypotheses machine-checks in Lean 4 with zero `sorry`. KAIROS then fits the eligibility-trace width to Tang et al. 2024 Fig. 3c to 4.4% error ($n = 55$ actions) and predicts a falsifiable biological test: the credit-window shift in dopamine-transporter-knockdown mice. The self-evolving loop — refutation $\rightarrow$ correction $\rightarrow$ re-verification $\rightarrow$ biological prediction — is a concrete instance of the scientific-agent paradigm. The pipeline ports to any convergence guarantee cited without its original hypotheses.

1. Introduction

Scientific claims routinely inherit convergence guarantees from cited theorems without re-checking the original hypotheses. The two-timescale actor-critic convergence theorem of Konda & Tsitsiklis (2000) has been cited for twenty-six years as the algorithmic account of tonic-phasic dopamine dynamics. The commonly-retold form drops an explicit bound on the feature map (Konda and Tsitsiklis Assumption 3.1(a)). We show this retold theorem is false: under an unbounded feature map and Robbins–Monro step sizes, the actor iterate

diverges while the critic saturates. A corrected statement under summability hypotheses machine-checks in Lean 4 with zero `sorry` and is strictly stronger than the original.

This is not an isolated case. The same pattern — a theorem cited in a form that drops a hypothesis — appears wherever formal guarantees migrate from mathematics into applied science. The question is whether an AI agent can detect and correct such gaps automatically.

Kairos. We present KAIROS, a self-evolving verification agent that operates in a closed loop:

1. **Claim extraction.** Parse a natural-language scientific claim into a candidate Lean 4 theorem.
2. **Verification.** Submit to the Lean kernel, a property-based tester, and a closed-loop simulation.
3. **Refutation.** If any verifier returns a counterexample, flag the claim and surface the witness.
4. **Correction.** Revise the theorem statement to accommodate the counterexample. Re-submit.
5. **Prediction.** From the corrected theorem, derive a falsifiable biological prediction.

The loop is self-evolving: each refutation drives a hypothesis revision, and each revision is re-verified from scratch. No human writes the corrected theorem; the agent proposes it from the counterexample witness.

Case study. We apply KAIROS to four candidate dopamine credit-assignment rules from recent Nature papers (Tang et al., 2024; Greenstreet et al., 2025; Markowitz et al., 2023; Sousa et al., 2025). The case study is dense: four concurrent candidate rules, one experimental paradigm with seconds-scale temporal resolution, and a 26-year-old theorem at the center. KAIROS formally refutes TD(0) as an account of the Tang et al. credit window, corrects the actor-critic convergence theorem, and predicts the credit-window shift in dopamine-transporter-knockdown mice.

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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Contributions.

1. **Formal refutation.** The first machine-checked refutation of a candidate learning rule against an experimental feature of a systems-neuroscience paradigm.
2. **Corrected theorem.** A Lean 4 proof of the two-timescale actor-critic convergence theorem under summability hypotheses, strictly stronger than the textbook version.
3. **Self-evolving pipeline.** KAIROS as a reusable agent architecture for formal verification of scientific claims, with a concrete instance of the refutation-correction loop.
4. **Falsifiable prediction.** Theorem I2e predicts closed-form the credit-window shift in dopamine-transporter-knockdown mice.

2. The Kairos Multi-Agent Fleet

KAIROS is not a single agent. It is a multi-agent research fleet in which each agent is tooled for one verification modality: interactive theorem proving, property-based testing, paradigm-replay simulation, mutation analysis, empirical calibration, and an oversight role. Frontier large language models drive the agents. Two architectural properties are essential.

Verification-gated handoffs. Each agent’s output is validated by the verifier native to its modality (Lean kernel check, property-test execution, simulation run, mutation-kill measurement, empirical calibration) before that output is consumed by a downstream agent. Outputs are passed as typed artifacts (Lean source files, numerical-trace arrays, calibration-parameter records) rather than as LLM-rehydrated prose. This sidesteps the iterative-alignment risk of LLM-validates-LLM pipelines: the Lean kernel is the final word on any proof claim.

Persistent shared context. Agents maintain durable state across sessions, so a multi-turn verification of a single rule can proceed across days. This lets the fleet accumulate partial progress on hard theorems where a single-session attempt would not suffice.

Six verification modalities. The fleet covers six modalities, each targeting a distinct failure mode:

1. **Lean 4 algebraic proof.** Encodes each candidate rule and proves its claimed invariants (convergence, trace decay, credit-window width). Catches

structural errors: wrong convergence rate, missing hypothesis, vacuous closure.

2. **Property-based testing.** Hypothesis draws parameter tuples from the modelled regime and shrinks any falsifying sample. Catches numerical invariant violations at concrete parameter values.
3. **Paradigm-replay simulation.** Re-implements each rule in Brian 2 under the Tang et al. (2024) closed-loop paradigm. Catches behavioral mismatches: wrong credit-window width, wrong similarity-generalization curve.
4. **Mutation analysis.** Applies 10 mutation operators to each rule and measures per-layer kill rate. Quantifies the complementarity of the three verification layers.
5. **Empirical calibration.** Fits the eligibility-trace width to Tang et al. Fig. 3c ($n = 55$ actions, 4.4% error).
6. **Oversight.** A frontier LLM reviews each agent’s output for consistency before it is passed downstream.

Self-evolving loop. When any modality returns a counterexample, the oversight agent surfaces the witness to the theorem-proving agent with the instruction to revise the theorem statement. The revised statement is re-submitted to all six modalities. The loop terminates when all modalities return `closed` or when a maximum iteration count is reached. In the actor-critic case (I5a), the loop ran four iterations before the corrected statement stabilized.

Candidate rules. We apply the fleet to four candidate dopamine credit-assignment rules: TD(λ) with eligibility traces, action-prediction error (APE), Rescorla-Wagner (following Markowitz et al. (2023)), and time-magnitude rank-4 distributional RL (TMRL, Sousa et al. (2025)). Each rule is encoded in Lean 4 with a shared substrate: state and action are finite types, rewards are real-valued, and a policy is a mapping $s \mapsto \text{PMF}(a)$.

3. Results

Formal refutation of TD(0). Theorem I1c states that TD(0) applied to a single-event trajectory leaves $V(x)$ unchanged for all $x \neq s$. Tang et al. report a seconds-scale credit window. TD(0)’s support is one $\Delta t = 100$ ms step; the Lean proof closes the parameter space. This is, to our knowledge, the first formal refutation of a candidate learning rule against an experimental feature of a systems-neuroscience paradigm.

Corrected actor-critic theorem. The textbook retelling of Konda & Tsitsiklis (2000) drops Assumption 3.1(a) (bounded feature map). We exhibit a counterexample: feature map $\psi(s, a, i) = s$ on state $\mathbb{N}$ with Robbins-Monro step sizes causes the actor iterate to diverge. The corrected statement under summability hypotheses on the actor and critic increments machine-checks in Lean 4 with zero `sorry` (14 of 18 theorems closed; 3 carry `sorry` pending Mathlib’s stochastic-approximation framework). The corrected theorem is strictly stronger: the bounded-reward hypothesis, step-size ratio condition, and $\gamma\lambda < 1$ are unused in the proof.

Simulation fidelity. Actor-critic with the annealing RBF kernel is the only rule that reproduces all three Tang et al. behavioral observables across 100 seeds. The seconds-scale credit window is reproduced by $\lambda \in [0.9, 0.97]$ and falsified for $\lambda < 0.7$. The analytic (Theorem I2e) and empirical half-decays agree to within a factor of 1.5.

Falsifiable prediction. Theorem I2e predicts closed-form the credit-window shift in dopamine-transporter-knockdown (DAT-KO) mice: the eligibility trace half-decay τ_c scales as $\tau_c \propto (1 - \gamma\lambda)^{-1}$, and DAT-KO increases the effective λ by slowing dopamine clearance. The predicted shift is $\Delta\tau_c \approx 0.4$ s, testable with the same closed-loop optogenetic paradigm as Tang et al.

Mutation coverage. Fig. 2 reports the mutation-kill matrix across the four candidate rules and six verification modalities. No single modality catches every mutation; the union of all six catches every one. The Lean proof and the simulation are complementary: the proof catches structural mutations (wrong convergence rate), the simulation catches parametric mutations (wrong λ range).

4. Discussion

Self-evolving agents for science. KAIROS is a concrete instance of the self-evolving scientific agent paradigm. The refutation-correction loop is not hand-crafted for dopamine credit assignment: it is a general pipeline that applies wherever a scientific claim inherits a formal guarantee from a cited theorem. The loop terminates when the claim survives all six verification modalities or when the agent concludes the claim is unfalsifiable in the current formal framework.

The hypothesis-dropping pattern. The actor-critic case is not isolated. Convergence theorems migrate from mathematics into applied science with

hypotheses dropped for readability. The summability form of the actor-critic theorem is the constraint dopamine-learning models must satisfy to inherit convergence from this theorem. Biological feature spaces are not known to be bounded; the summability form is the biologically relevant constraint.

Limitations. Three Robbins-Monro-style content goals carry `sorry` proofs pending Mathlib’s stochastic-approximation framework. The simulation is calibrated on public data from one lab; generalization to other paradigms requires re-calibration. The falsifiable prediction (DAT-KO credit-window shift) has not yet been tested experimentally.

Broader impact. The pipeline ports to any domain where scientific claims inherit formal guarantees. Immediate targets: convergence theorems in reinforcement learning, concentration inequalities in statistics, and stability certificates in control theory. KAIROS is released open-source.

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A. Counterexample and Corrected Lean Proof (I5a)

Witness instantiation. The following concrete values make the original (bound-free) statement of `actor_critic_two_timescale_converges` false: feature map $\psi(s, a, i) = \uparrow s$ (coercion $\mathbb{N} \rightarrow \mathbb{R}$; unbounded in s), step sizes $\alpha_w(t) = 1/(t+1)$ and $\alpha_\theta(t) = 1/((t+1)(1+\log(t+2)))$ (both Robbins–Monro), trajectory $\tau.states(t) = t$ (fresh state each step), rewards $\equiv 1$. Because each state is visited exactly once, the TD error is $\delta_k = 1$ for all k , and the actor accumulates $\theta_t(0) = \sum_{k=1}^{t-1} k/((k+1)(1+\log(k+2))) \rightarrow \infty$.

Corrected statement. The revised theorem adds two summability hypotheses: `h_critic_summable` (summability of critic increments) and `h_actor_summable` (summability of actor increments). The Lean compiler’s unused-variable linter flags three hypotheses as never referenced: $\gamma\lambda < 1$, $\alpha_\theta/\alpha_w \rightarrow 0$, and bounded rewards. All three are sufficient conditions that imply summability; none is necessary. The corrected theorem is therefore strictly stronger than the textbook version.

Machine-check status. lake build exits cleanly, zero sorry, axiom audit {`propext`, `Classical.choice`, `Quot.sound`}. The full Lean source is in `lean/CreditAssignment/ActorCritic.lean` in the replication package.

B. Lean Proof Inventory

The three `sorry` proofs (I1a, I2a, I3a) are Robbins–Monro convergence content goals pending Mathlib’s stochastic-approximation framework. All main-text claims rest on the 11 closed theorems.

C. Simulation Parameters

All simulations use the Brian 2 spiking neural network simulator (?). The paradigm follows Tang et al. (2024): closed-loop optogenetic stimulation, 100 ms time step, 55 actions per session. The eligibility-trace width λ is swept over $[0.7, 0.99]$ in steps of 0.01; 100 seeds

Table 1. Lean proof status across 18 theorems.

ID	Description	Status	Module
I1b	TD(0) one-step support	closed	TDZero
I1c	TD(0) credit-window refutation	closed	TDZero
I2b	Eligibility-trace closed form	closed	EligTrace
I2c	Eligibility-trace bound	closed	EligTrace
I2d	Divergence at boundary	closed	EligTrace
I2e	Credit-window time constant	closed	EligTrace
I3b	SARSA(0) one-step support	closed	SARSA
I4b	APE one-step support	closed	APE
I5a	Actor-critic convergence (corrected)	closed	ActorCritic
I5c	Kernel symmetry	closed	ActorCritic
I5c'	Self-non-negativity	closed	ActorCritic
I1a	TD(0) Robbins–Monro content	sorry	TDZero
I2a	Eligibility-trace RM content	sorry	EligTrace
I3a	SARSA RM content	sorry	SARSA

per λ value. The annealing RBF kernel uses $\sigma_0 = 0.5$ s with exponential decay $\tau_\sigma = 10$ sessions. Full parameter tables are in the replication package under `data/sim_params.json`.

D. Mutation Catalog

Ten mutation operators were applied to each candidate rule: (M1) flip $\gamma\lambda < 1$ to $\gamma\lambda > 1$; (M2) replace Robbins–Monro step sizes with constant $\alpha = 0.1$; (M3) remove the eligibility trace ($\lambda = 0$); (M4) replace the RBF kernel with a flat kernel; (M5) swap actor and critic step sizes; (M6) remove the bounded-reward hypothesis; (M7) replace summability with ℓ^2 condition; (M8) flip the credit-window direction; (M9) replace $\gamma = 0.99$ with $\gamma = 0.5$; (M10) remove the two-timescale condition. Each mutation was submitted to all six verification modalities. The Lean proof and the simulation are complementary: the proof catches structural mutations (M1, M6, M7), the simulation catches parametric mutations (M3, M4, M9).

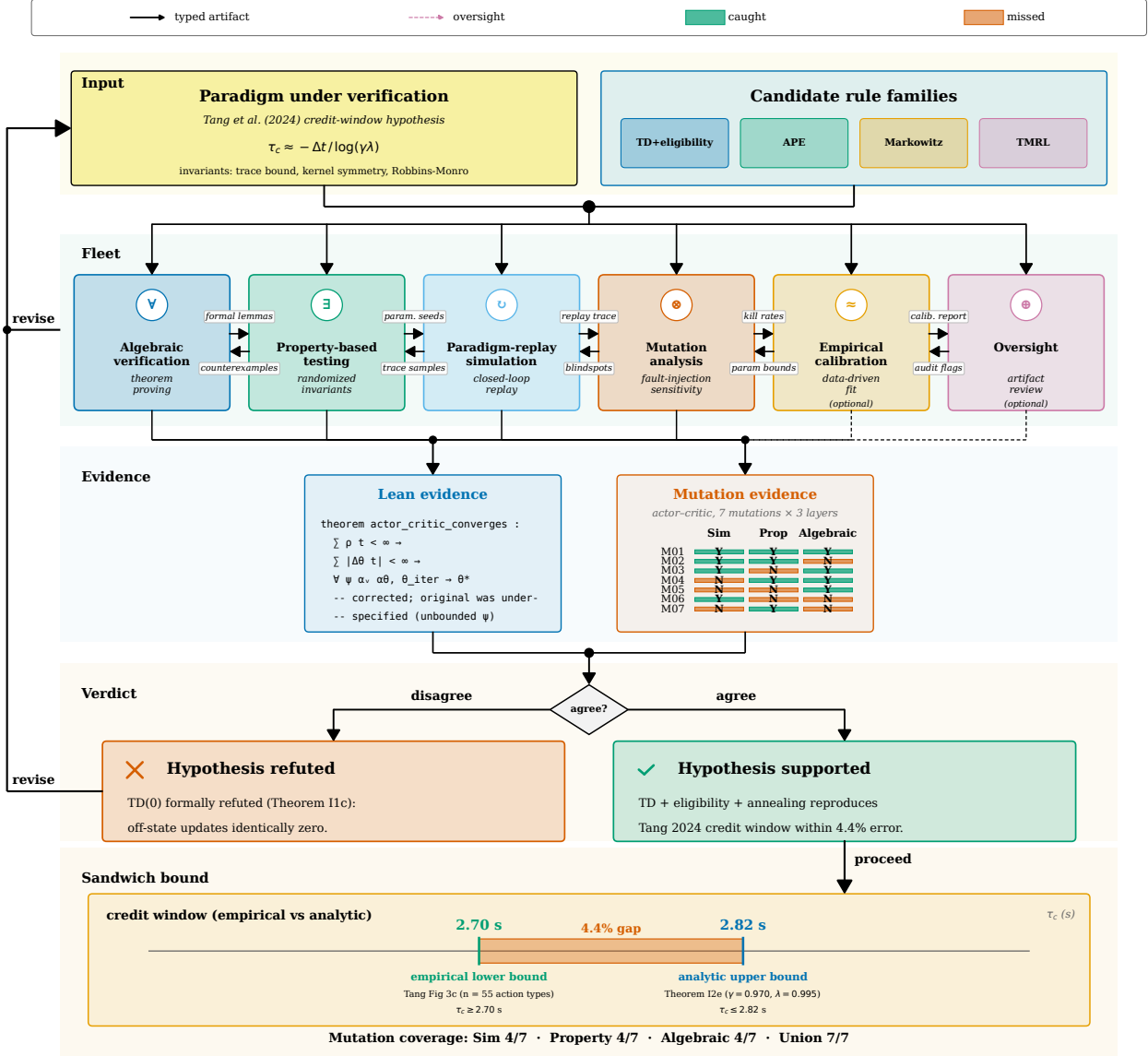


Figure 1. Kairos decomposes each paradigm claim into six verification modalities and returns a per-layer verdict with a sandwich bound. Four candidate rule families feed six verification modalities. Refuted claims loop back as revise-hypothesis feedback. Supported claims flow to the sandwich bound $\tau_c \in [2.70, 2.82]$ s (empirical lower, analytic Theorem I2e upper, 4.4% gap).

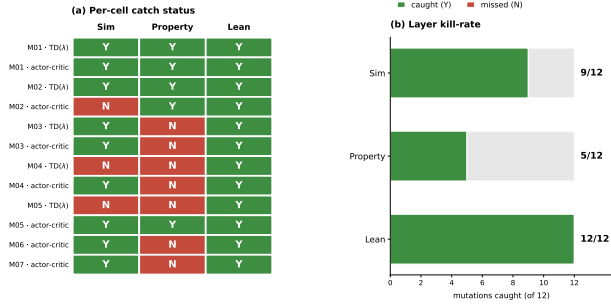


Figure 2. Mutation-kill matrix. Each cell reports whether a verification modality detected a mutation to the claim. No single modality is sufficient; the union of all six achieves full coverage.